

Students Grade Data & Analysis										
Student ID	Full Name	Math	Science	English	Average	Grade	Status (Math & Science >80)	First Name	Name (UPPER)	Name (LOWER)
S001	Rahul Sharma	85	78	92	85.00	B	Needs Improvement	Rahul	RAHUL SHARMA	rahul sharma
S002	Priya Patel	92	88	85	88.33	B	Excellent	Priya	PRIYA PATEL	priya patel
S003	Amit Kumar	45	52	60	52.33	F	Needs Improvement	Amit	AMIT KUMAR	amit kumar
S004	Sneha Reddy	78	81	75	78.00	C	Needs Improvement	Sneha	SNEHA REDDY	sneha reddy
S005	Vikram Singh	95	90	88	91.00	A	Excellent	Vikram	VIKRAM SINGH	vikram singh
S006	Ananya Gupta	62	55	70	62.33	D	Needs Improvement	Ananya	ANANYA GUPTA	ananya gupta
S007	Rohan Mehta	88	85	90	87.67	B	Excellent	Rohan	ROHAN MEHTA	rohan mehta
S008	Kavita Joshi	55	48	52	51.67	F	Needs Improvement	Kavita	KAVITA JOSHI	kavita joshi
S009	Arjun Nair	73	79	68	73.33	C	Needs Improvement	Arjun	ARJUN NAIR	arjun nair
S010	Meera Iyer	91	94	89	91.33	A	Excellent	Meera	MEERA IYER	meera iyer
S011	Siddharth Rao	40	35	45	40.00	F	Needs Improvement	Siddharth	SIDDHARTH RAO	siddharth rao
S012	Divya Krishnan	82	87	80	83.00	B	Excellent	Divya	DIVYA KRISHNAN	divya krishnan
Summary Analysis (COUNTIFS / SUMIFS / AVERAGEIFS / FILTER)										
Metric	Formula / Result	Value								
Students with Math > 80	=COUNTIFS(B2:B12,>80)	10								
Students with Average > 80	=COUNTIFS(E2:E12,>80)	9								
Avg Math score (Students with Math > 80)	=AVERAGEIFS(B2:B12,B2:B12,>80)	82.88888889								
Students with Math > 80 and Average > 80	=COUNTIFS(B2:B12,>80,E2:E12,>80)	5								
FILTER: Students with Average > 80 (Top Performers)										
Student ID	Full Name	Math	Science	Average						
S002	Priya Patel	92	88	88.33						
S005	Vikram Singh	95	90	91.00						
S007	Rohan Mehta	88	85	87.67						
S010	Meera Iyer	91	94	91.33						
S012	Divya Krishnan	82	87	83.00						
VLOOKUP Demo: Find Student Name by ID										
Enter Student ID	S006									
Student Name (VLOOKUP)	Ananya Gupta									
Math Score (VLOOKUP)	62									

Sales Data & Lookup / Aggregation Analysis

Product Master Table									
Product Code	Product Name	Unit Price	Category						
P101	Laptop Pro 15	₹85,000	Electronics						
P102	Wireless Mouse	₹1,200	Accessories						
P103	USB-C Hub	₹2,500	Accessories						
P104	Monitor 27"	₹18,500	Electronics						
P105	Keyboard Mecha	₹4,500	Accessories						
P106	Webcam HD	₹3,200	Electronics						
P107	Laptop Stand	₹1,800	Accessories						
P108	Noise Cancelling	₹9,500	Electronics						
Sales Transactions									
Sale ID	Date	Product Code	Product Name (VLOOKUP)	Unit Price (VLOOKUP)	Qty	Region	Salesperson	Total Sales	Discount Eligible?
SL001	15-Jan-2025	P101	Laptop Pro 15	₹85,000	2	North	Ravi	₹170,000	Yes
SL002	18-Jan-2025	P102	Wireless Mouse	₹1,200	5	South	Priya	₹6,000	Yes
SL003	03-Feb-2025	P104	Monitor 27"	₹18,500	1	West	Amit	₹18,500	Yes
SL004	10-Feb-2025	P103	USB-C Hub	₹2,500	3	North	Ravi	₹7,500	No
SL005	22-Feb-2025	P108	Noise Cancelling Headphones	₹9,500	2	East	Sneha	₹19,000	No
SL006	05-Mar-2025	P105	Keyboard Mechanical	₹4,500	4	South	Priya	₹18,000	No
SL007	12-Mar-2025	P101	Laptop Pro 15	₹85,000	1	West	Amit	₹85,000	Yes
SL008	20-Mar-2025	P106	Webcam HD	₹3,200	2	North	Ravi	₹6,400	No
SL009	02-Apr-2025	P107	Laptop Stand	₹1,800	6	East	Sneha	₹10,800	Yes
SL010	15-Apr-2025	P102	Wireless Mouse	₹1,200	8	South	Priya	₹9,600	Yes
SL011	01-May-2025	P104	Monitor 27"	₹18,500	2	North	Ravi	₹37,000	Yes
SL012	18-May-2025	P108	Noise Cancelling Headphones	₹9,500	1	West	Amit	₹9,500	No
Discount Calculation (Nested IF)									
Product Code	Unit Price	Discount %	Discounted Price						
P101	₹85,000	15%	₹72,250.00						
P102	₹1,200	0%	₹1,200.00						
P103	₹2,500	0%	₹2,500.00						
P104	₹18,500	10%	₹16,650.00						
P105	₹4,500	0%	₹4,500.00						
P106	₹3,200	0%	₹3,200.00						
P107	₹1,800	0%	₹1,800.00						
P108	₹9,500	5%	₹9,025.00						
Aggregation with SUMIFS / COUNTIFS / AVERAGEIFS									
Metric	Criteria	Result							
Total Sales - Nort Region = North		₹220,900							
Total Sales - Prod Product = P101		₹255,000							
Total Sales - Nort Region=North AND		₹170,000							
Count of Sales - S Region = South		3							
Avg Sale Value - E Using AVERAGEIF		₹33,108.33							
XMATCH Function Demo									
Find position of P105									
Position in Product	5								
Product Name at Keyboard Mechanical									
INDEX + MATCH: Sales by Salesperson									
Select Salesperson	Ravi								
First matching Sale ID	SL001								
Total Sales by this person	₹220,900								
INDIRECT Function Demo									
Sheet Name (type)	Sales Data								
Cell Reference:	C5								
Value via INDIRECT	₹85,000								
OFFSET Function Demo - Dynamic Range									

Starting from first	5							
Dynamic SUM us	₹221,000							

Employee Data & Advanced Lookups / Date Functions

Emp ID	Full Name	Department	Date of Joining	Date of Birth	Salary	Years of Service	Age	First Name	Salary (XLOOKUP)	Status
E001	Rajesh Khanna	Sales	15-Mar-2018	20-Jun-1985	₹65,000	8	41	Rajesh	₹65,000	Senior
E002	Sunita Agarwal	HR	01-Jul-2019	05-Nov-1990	₹55,000	7	35	Sunita	₹55,000	Senior
E003	Mohit Verma	IT	10-Jan-2017	14-Feb-1988	₹85,000	9	38	Mohit	₹85,000	Senior
E004	Pooja Desai	Finance	22-Sep-2020	30-Aug-1992	₹70,000	6	34	Pooja	₹70,000	Senior
E005	Karan Malhotra	Sales	05-May-2016	12-Dec-1983	₹72,000	10	42	Karan	₹72,000	Senior
E006	Neha Kapoor	Marketing	18-Feb-2021	25-Apr-1995	₹48,000	5	31	Neha	₹48,000	Senior
E007	Aakash Jain	IT	30-Nov-2019	08-Sep-1991	₹78,000	6	35	Aakash	₹78,000	Senior
E008	Ritu Sharma	HR	12-Apr-2022	17-Jan-1994	₹52,000	4	32	Ritu	₹52,000	Mid-Level
E009	Vishal Reddy	Finance	08-Aug-2015	03-Jul-1980	₹95,000	11	46	Vishal	₹95,000	Senior
E010	Anjali Nair	Marketing	25-Jun-2020	19-Oct-1993	₹58,000	6	32	Anjali	₹58,000	Senior
XLOOKUP Demo (implemented with INDEX/MATCH for compatibility)										
XLOOKUP Demo: Fetch Employee Salary Dynamically										
Enter Emp ID →	E003									
Employee Name	Mohit Verma									
Department (XLOOKUP)	IT									
Salary (XLOOKUP)	₹85,000									
Date of Joining (XLOOKUP)	10-Jan-2017									
Date Difference Demo										
Start Date:	01-Jan-2024									
End Date:	02-Oct-2025									
Difference in Day	640									
Difference in Year	1.75									
Math Functions: ROUND / CEILING / FLOOR										
Original Salary	ROUND to nearest	CEILING to next \$	FLOOR to previous							
₹65,000	₹65,000	₹65,000	₹65,000							
₹55,000	₹55,000	₹55,000	₹55,000							
₹85,000	₹85,000	₹85,000	₹85,000							
₹70,000	₹70,000	₹70,000	₹70,000							
₹72,000	₹72,000	₹75,000	₹70,000							
INDEX + MATCH: Flexible Employee Lookup (no column reference)										
Lookup Emp ID:	E007									
Department (INDEX/MATCH)	IT									
Salary (INDEX/MATCH)	₹78,000									
Years of Service (INDEX/MATCH)	6									
Relative vs Absolute References Demo										
Tax Rate (Absolute reference \$B\$46):										
	10%									
Emp ID	Salary	Tax (using \$B\$46)	Net Salary							
E001	₹65,000	₹6,500.00	₹58,500.00							
E002	₹55,000	₹5,500.00	₹49,500.00							
E003	₹85,000	₹8,500.00	₹76,500.00							
E004	₹70,000	₹7,000.00	₹63,000.00							
E005	₹72,000	₹7,200.00	₹64,800.00							